Project title: Detection of helicobacter pylori in gastric biopsy
Supervisor: Prof. Dr. Magda Gregorová
Student team: Harsha Sathish, Rohan Sanjay Patil, Vidya Padmanabha

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1. Introduction

Problem Statement

Helicobacter pylori (H. pylori) is a tiny, spiral-shaped bacteria that lives in the stomach and affects intensively. This bacterium can cause serious health problems—including stomach ulcers, chronic inflammation, and stomach cancer. Health organizations, including the World Health Organization, have officially classified H. pylori as a cancer-causing agent, making it a significant global health concern.

Why This Matters

Catching H. pylori early can prevent these serious health complications. But here's the catch: detecting it isn't easy. Right now, doctors examine stomach tissue samples under a microscope—a process that's slow, tiring, and inconsistent. Because H. pylori is incredibly small and can look different in different samples, even trained specialists sometimes miss it. While alternative methods such as immunohistochemistry offer improved accuracy, their high cost and technical demands limit their widespread use.

Proposed Solution

Our project aims to make H. pylori detection faster, more reliable, and accessible to everyone. By leveraging modern technology and digital tools, we want to help doctors spot this bacteria more consistently in stomach tissue samples. The goal is simple: better detection means earlier treatment.

2. Literature review

The primary challenge our project addresses is the detection of tiny, few-pixel long H. pylori bacteria within massive, gigapixel sized Whole Slide Images (WSIs). This "needle in a haystack" problem makes manual checks by pathologists a time-consuming and labor-intensive task. Based on the research by Klein et al. we can conclude that a "coarse-to-fine" pipeline is the most practical approach. The researchers first used standard image processing to find "hot spot" regions (gastric glands) and then applied a CNN classifier to patches from these regions. This two-stage idea is impressive, but it's simple method for finding glands is a major bottleneck of the research.

This limitation was evident in our Seniors' research paper, which adopted a similar strategy. Their "template matching" technique for finding candidate glands was unreliable, missing almost 20% of the true positive regions before the AI model even had a chance to analyze them. This proves that the first step of accurately identifying all the relevant gland regions requires a more intelligent and robust solution, such as the U-Net segmentation model we propose.

The seniors' work also explored a more advanced object detection pipeline, similar to our current plan, using NuClick to mask/define necessary glands and a YOLO model/CNN model to find the bacteria. However, their results were constrained by a different, more fundamental problem of poor annotations. Their paper explicitly states their ground-truth labels were "not entirely precise" and "limited" with many true positives likely mislabeled as negative. This unreliable data makes it extremely difficult to train a high-performance detector, which is reflected in their final results.

This "imprecise label" problem is a known challenge in digital pathology. Yang et al. tackled this issue by pioneering a "weakly supervised multi-task learning" framework. They recognized that perfect, pixel-level labels are impractical to obtain. Their solution was to train a single model using two imperfect targets simultaneously: one "broad" target (where false positives are possible) and one "fine-grained" target (where false negatives might show up). This forces the model to find a robust compromise. Their work shows a clear path forward, suggesting that our project can succeed by adopting similar weak-supervision techniques to overcome the exact annotation challenges that limited previous efforts if we fail to obtain the precise annotations after our classification/detection and re-annotation(by Philipp) loop.

3. Research Question, Hypothesis or Aim and Objectives

- Handle massive image files efficiently – Gastric biopsy slides are with billions of pixels, creating significant computational challenges that we need to solve.
- Build a prototype of a detection system that can identify H. pylori in stomach tissue samples quickly and accurately.
- Speed up diagnosis by creating a faster alternative to the current manual process, helping doctors get results sooner.

4. Research design/methodology

The proposed approach involves object detection method in a hierarchical fashion.

1. First Item Pre-processing and Background Filtration WSIs contain a lot of irrelevant data like white spaces and background regions (about 60-70%) that slow down processing and make the model less efficient. Also, the lighting in HE-stained biopsy slides varies, which creates problems for detection. To fix this, we convert WSI patches to HSV (Hue, Saturation, Value) color space because it handles lighting changes better. Then we use the OTSU thresholding method to find and remove background regions. This removes 60-70% of unwanted data while keeping the important tissue parts we need for analysis.
2. Tissue Segmentation (Gastric Gland Isolation) H. pylori bacteria mostly live in gastric glands, so looking at the entire tissue would include parts that don't matter and make detection less accurate. We need to segment and isolate just the gastric gland regions where H. pylori is likely to be found. We'll use QuPath's built-in pixel classifier to segment gastric glands because it's faster than NuClick and does similar work. If QuPath doesn't work well enough, we'll switch to the NuClick/U-Net model as a backup option.
3. Sliding Window Patch Extraction After segmentation, the images might still be too big to feed directly into the model. Deep learning models need fixed-size patches, and we also need to avoid losing information where patches meet. So we divide the high-resolution images into 320×320 pixel patches using a sliding window with 20% overlap between patches. This patch size gives the model enough context to understand the tissue structure while keeping things manageable for processing.

4. Active Learning for Missing Annotations Our dataset has a lot of missing annotations, which makes it hard to train the model properly. Annotating everything manually takes too much time. To solve this, we'll use active learning. First, we train a baseline YOLO model using the samples that are already annotated (both positive and negative). Then we use this trained model to create annotations for the unlabeled samples in our dataset, which helps us expand our training data without doing everything manually.

5. Object Detection (H. pylori Localization) For diagnosis, we need to know exactly where the bacteria are located so pathologists can check the detections. The YOLO model is good for this because it can detect objects in real-time, which works well in clinical settings. We apply YOLO to the segmented and patched images, and it gives us bounding boxes with confidence scores showing where each bacteria is located in the tissue.

6. Class Imbalance Mitigation Our dataset has a big problem: there's a 1:100 imbalance between positive samples (with bacteria) and negative samples (without bacteria). If we train normally, the model will just learn to predict negative most of the time, which is bad because finding bacteria is what actually matters. We use three techniques to fix this. First, we implement Asymmetrical Focal Loss, which makes the model focus more on hard examples instead of easy negatives. Second, we use aggressive data augmentation with geometric changes and color jitter to create more variety in our positive samples and simulate how HE stains can look different. Third, we use batch-aware oversampling to make sure every training batch has at least some positive examples so the model doesn't forget about them.

7. Coordinate Transformation (WSI-level Localization) When YOLO finds bacteria, it gives coordinates that are only relative to the small patch it looked at. But doctors need to know where the bacteria are on the original full-size slide so they can find and verify them. To do this, we save the position of each patch's top-left corner relative to the original WSI. Then we use these saved positions to convert YOLO's patch coordinates into coordinates on the full WSI. This way, the final bounding boxes can be shown directly on the original whole slide images.

5. Project plan

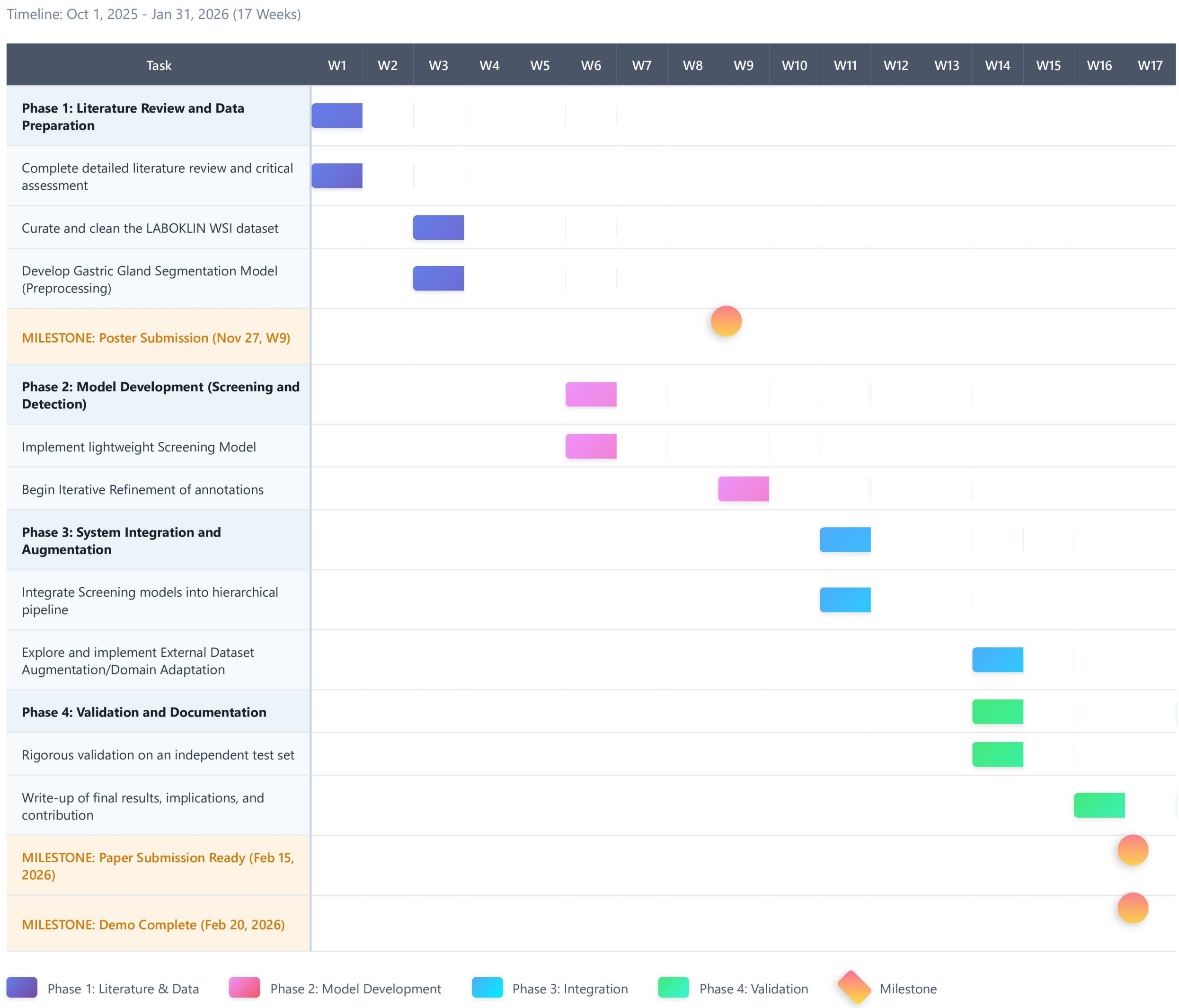

Figure 1: Detailed Project Plan

https://github.com/users/hxrshx/projects/1/views/4?visibleFields=

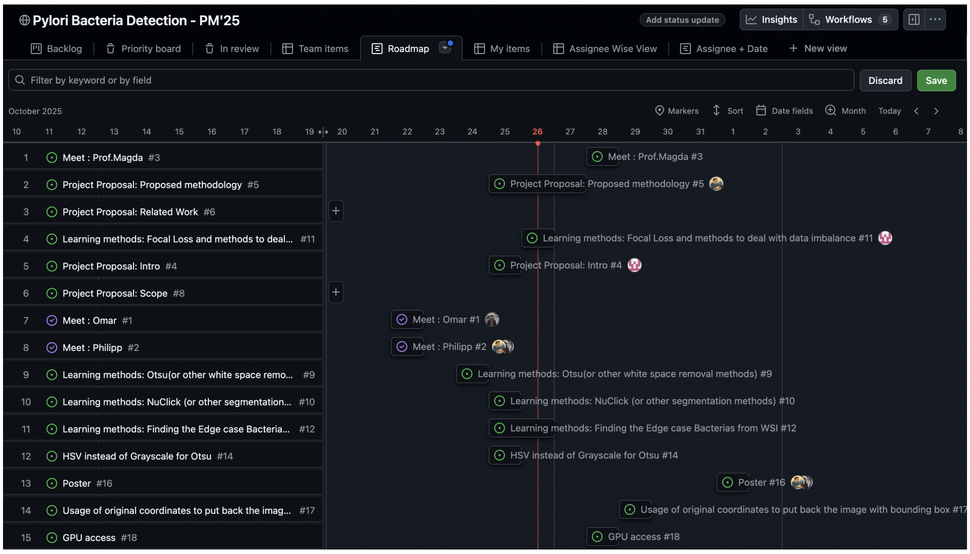

Figure 2: Current Task based Project Plan

6. Expected Outcome/ Implications and Contribution to the knowledge

The end result will be a prototype system demonstrating automated H. pylori detection for pathologists. The prototype will feature a simple web interface where pathologists can upload a Whole Slide Image (WSI). Our system will then analyze the WSI to identify and rank the 20-30 most likely areas containing H. pylori bacteria. The results will be presented as color-coded bounding boxes overlaid on the WSI: 10-15 red boxes highlighting the highest confidence "hot spots" where bacteria are most likely present, and 10-15 yellow boxes showing the next highest confidence regions. This approach changes how pathologists review slides - instead of scanning the entire slide manually, they can focus immediately on the most critical areas (red boxes first, then yellow boxes) to quickly confirm a positive case or confidently clear a slide as negative.

Project Deliverables The project will be presented through multiple deliverables:

- Academic Poster: A visual poster summarizing the project's objectives, methodology, results, and conclusions will be created for presentation.
- Research Paper: A detailed technical paper documenting the methodology, experimental results, and findings will be prepared.
- Live Demonstration: An interactive demo of the web-based prototype will be conducted to showcase the system's functionality.

References

Klein, Sebastian, Jacob Gildenblat, Michaele Angelika Ihle, Sabine Merkelbach-Bruse, Ka-Won Noh, Martin Peifer, Alexander Quaas, and Reinhard Büttner (2020). "Deep learning for sensitive detection of Helicobacter Pylori in gastric biopsies". en. In: *BMC Gastroenterology* 20.1, p. 417.

Yang, Yongquan, Yiming Yang, Yong Yuan, Jiayi Zheng, and Zheng Zhongxi (2020). "Detecting helicobacter pylori in whole slide images via weakly supervised multi-task learning". en. In: *Multimedia Tools and Applications* 79.35-36, pp. 26787–26815.